

Topic to discuss

- Numerical differentiation for unequal interval
- Newton divided difference derivative
- Numerical Problem.

Newton divided difference formula,

$$y = f(x) = y_0 + (x-x_0) \Delta y_0 + (x-x_0)(x-x_1) \Delta^2 y_0 + \\ (x-x_0)(x-x_1)(x-x_2) \Delta^3 y_0 + \dots$$

$$\Rightarrow y = f(x) = y_0 + (x-x_0) \Delta y_0 + (x^2 - x x_1 - x x_0 + x_1 x_0) \Delta^2 y_0 + \\ + (x^3 - x^2 x_1 - x^2 x_0 + x x_1 x_0 - x^2 x_2 + x x_1 x_2 + x x_0 x_2 - x_0 x_1 x_2) \Delta^3 y_0$$

first order derivative w.r.t x .

$$y' = f'(x) = \Delta y_0 + (2x - x_1 - x_0) \Delta^2 y_0 + \\ (3x^2 - 2x x_1 - 2x x_0 + x_1 x_0 - 2x x_2 + x_1 x_2 + x_0 x_2) \Delta^3 y_0$$

$$y' = f'(x) = \Delta y_0 + (2x - x_0 - x_1) \Delta^2 y_0 + \\ (3x^2 - 2x(x_0 + x_1 + x_2) + x_0 x_1 + x_1 x_2 + x_2 x_0) \Delta^3 y_0$$

Second order derivative w.r.t x .

$$\therefore y' = f'(x) = \Delta y_0 + (2x - x_1 - x_0) \Delta^2 y_0 + \\ (3x^2 - 2xx_1 - 2xx_0 + x_1x_0 - 2xx_2 + x_1x_2 + x_0x_2) \Delta^3 y_0 + \dots$$

$$\therefore y'' = f''(x) = 2 \Delta^2 y_0 + (6x - 2x_1 - 2x_0 - 2x_2) \Delta^3 y_0 + \dots$$

$$y'' = 2 \Delta^2 y_0 + (6x - 2(x_0 + x_1 + x_2)) \Delta^3 y_0 + \dots$$

Numerical Problem

Q:- find y' and y'' at $x=3$ for given table,

x	1	2	4	7
y	2	3	5	10

Solution: Newton divided difference table,

x	y	Δy	$\Delta^2 y$	$\Delta^3 y_0$
$x_0 = 1$	$y_0 = 2$	$\Delta y_0 = 1$	$\Delta^2 y_0 = 0$	$\Delta^3 y_0 = \frac{1}{45}$
$x_1 = 2$	$y_1 = 3$	$\Delta y_1 = 1$	$\Delta^2 y_0 = \frac{2}{15}$	
$x_2 = 4$	$y_2 = 5$	$\Delta y_2 = \frac{5}{3}$		
$x_3 = 7$	$y_3 = 10$			

Now, using y' formula,

$$y' = f'(x) = \Delta y_0 + (2x - x_1 - x_0) \Delta^2 y_0 + (3x^2 - 2x(x_0 + x_1 + x_2) + x_0 x_1 + x_1 x_2 + x_2 x_0) \Delta^3 y_0$$

$$= 1 + (2 \times 3 - 2 - 1) \times 0 + (3 \times 3^2 - 2 \times 3(1 + 2 + 4) + 1 \times 2 + 2 \times 4 + 4 \times 1) \times \frac{1}{45}$$

$$= 1 + 0 + (27 - 6 \times 7 + 2 + 8 + 4) \frac{1}{45}$$

$$= 1 - \frac{1}{45}$$

$$= \frac{44}{45}$$

$$\therefore y' \text{ at } x=3 \text{ is } \frac{44}{45}$$

Now find y'' ,

$$y'' = 2\Delta^2 y_0 + (6x - 2(x_0 + x_1 + x_2))\Delta^3 y_0$$

$$= 2 \times 0 + (6 \times 3 - 2(1 + 2 + 4)) \frac{1}{45}$$

$$= (18 - 14) \frac{1}{45}$$

$$= \frac{4}{45}$$

$\therefore y''$ at $x=3$ is $\frac{4}{45}$ Ans.

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